

THE STRUCTURE OF MATHEMATICAL REALITY

Phase II — Chapter 5

Symmetry and Invariance: Why Some Operations Are Powerful

Exercises

Self-Study Curriculum

1. In PCA, you find the eigenvectors of the covariance matrix. These eigenvectors are the directions along which the data varies independently. In what sense is PCA discovering a *symmetry* of the data? What is the invariant? What “transformation” is the data symmetric under (approximately)?

2. Fourier analysis decomposes a signal into sinusoidal components. The Fourier transform is enormously powerful for certain problems and useless for others. In terms of symmetry: what symmetry does Fourier analysis exploit? (Hint: think about translation.) For what kind of signal does this symmetry not hold?

3. Consider the operation of sorting a list. A sorted list has a symmetry that an unsorted list does not. What is that symmetry? How does this symmetry make subsequent operations (searching, merging) more efficient?

4. A physical system has rotational symmetry. Without knowing anything else about the system,

what can you immediately predict about its behavior, using Noether's theorem?

5. Consider a problem you have solved by brute force (a grid search, an exhaustive enumeration, a combinatorial optimization). Ask: was the brute force a failure to spot a symmetry, or a genuine absence of exploitable structure? What evidence would distinguish the two? (You will not always be able to tell, recognizing *that* is the point.)
